

CricMetrics Pro: Complete File-by-File Repository Architecture Guide

Executive Scope: This document delivers an exhaustive, component-by-component architectural reference for every file and directory in the **CricMetrics Pro** codebase. For every single file, this specification details **WHAT** the file is, **HOW** it works internally, **WHEN** it is executed or utilized, and **WHY** it is necessary in the organizational architecture.

1. Repository Directory Structure & File Hierarchy

```
/Users/laxmanpatel/Desktop/ML
|-- app.py                               # Production Streamlit 6-in-1 War Room Web Dashboard
|-- requirements.txt                     # Production Python dependencies and pinned versions
|-- README.md                           # Public GitHub repository documentation and executive overview
|-- FINAL_REPORT.md                     # Comprehensive academic/organizational defense dossier
|-- .gitignore                           # Git exclusions (ignoring heavy raw data, preserving models & clean CSVs)
|-- .streamlit/
|   |-- config.toml                      # Streamlit server config, brand gold theme tokens, error settings
|-- data/
|   |-- processed/
|   |   |-- cricket_players_clean.csv    # Qualified 619-player analytical dataset with 27 engineered features
|   |-- raw/
|       |-- matches_2008_2024.csv        # 1,095 IPL match metadata (venues, teams, results, MoM awards)
|       |-- deliveries_2008_2024.csv     # 260,920 ball-by-ball deliveries (17 seasons, 69MB local dataset)
|-- models/
|   |-- scaler.joblib                    # Fitted StandardScaler object for 29 normalized features
|   |-- random_forest_regressor.joblib   # Production continuous rating model (R2 = 0.9789, RMSE = 1.25)
|   |-- ridge_regression.joblib           # L2 regularized linear benchmark model (R2 = 0.9491)
|   |-- linear_regression.joblib          # Ordinary Least Squares parametric baseline (R2 = 0.9490)
|   |-- polynomial_regression.joblib      # Degree-2 polynomial interaction model (R2 = 0.9112)
|   |-- mlp_regressor.joblib              # Deep neural network regression benchmark (R2 = 0.5495)
|   |-- knn_classifier.joblib             # Top talent tier classifier (95.16% Test Accuracy)
|   |-- random_forest_classifier.joblib   # Production tier classifier & Gini feature importance engine (94.35%)
|   |-- logistic_regression.joblib        # Multinomial softmax tier classification baseline (87.90%)
|   |-- decision_tree_classifier.joblib   # Rule-based CART tier classifier (87.90%)
|   |-- mlp_classifier.joblib             # Deep neural network tier classifier (92.74%)
|   |-- kmeans_model.joblib               # K-Means clustering model (k=5 tactical archetypes)
|   |-- pca_model.joblib                  # Principal Component Analysis model (2D projection, 60.8% variance)
|   |-- feature_metadata.json             # Feature names, schemas, categorical encodings, tier definitions
|   |-- metrics_summary.json              # Complete benchmark evaluation metrics and confusion matrices
|-- notebooks/
|   |-- cricket_player_performance_analysis.ipynb # 22-cell end-to-end data science study & interactive report
|-- src/
|   |-- cricket_data_pipeline.py          # Raw delivery aggregation & multi-skill feature engineering
|   |-- train_cricket_models.py           # Model training, 5-fold cross-validation & artifact serialization
|   |-- create_cricket_notebook.py        # Programmatic JSON generator for reproducible Jupyter notebook
|   |-- generate_all_docs.py              # Automated PDF documentation generator and builder
|-- docs/
|   |-- MACHINE_LEARNING_GUIDE.md & .pdf  # In-depth guide to all ML topics (What, When, Why, How)
|   |-- PROJECT_ARCHITECTURE_GUIDE.md & .pdf # File-by-file architectural blueprint (This document)
```

2. File-by-File Technical Deep Dive

2.1 User Interface & Application Layer

`app.py` — Production Streamlit 6-in-1 War Room Dashboard

WHAT The primary web application serving as the interactive decision-support war room for franchise executives, coaches, and sports data analysts (990+ lines of Python).

HOW Uses Streamlit, Plotly Express/Graph Objects, and Pandas. Utilizes `@st.cache_data` and `@st.cache_resource` to load dataset and 13 ML models into RAM. Organizes intelligence across 6 modules: *War Room & Roster Intel*, *Telemetry & Factor Impact* (radar charts and correlations), *AI Rating & Auction Valuation* (sliders, presets, purse calculator), *Tactical Archetypes & 2D Map* (PCA coordinates and K-Means clusters), *What-If Simulator*, and *Model Benchmarks & Defense*.

WHEN Continuously executed in real-time when accessed locally (`localhost:8501`) or hosted on Streamlit Community Cloud.

WHY Bridges complex machine learning algorithms and executive decision-makers. Franchise directors require real-time visual scenario analysis rather than command-line outputs.

`.streamlit/config.toml` — Theme & Server Configuration

WHAT Streamlit configuration file defining visual design tokens, server ports, and client error handling.

HOW Configures `primaryColor` to franchise gold (`#F59E0B`), `backgroundColor` to dark slate (`#0F172A`), `secondaryBackgroundColor` to navy (`#1E293B`), and sets `showErrorDetails = true`.

WHEN Automatically read by Streamlit at server initialization before app script execution.

WHY Enforces consistent enterprise dark-mode branding across all devices regardless of user browser defaults.

2.2 Data Ingestion & Pipeline Layer

`src/cricket_data_pipeline.py` — Delivery Aggregator & Feature Engine

WHAT Data ingestion and transformation script aggregating 260,920 raw deliveries into 619 qualified player profiles with 27 engineered metrics.

HOW Parses `deliveries_2008_2024.csv` and `matches_2008_2024.csv`. Calculates batting averages, strike rates, death overs (16–20) strike rates, bowling economies, dot ball percentages, and joins Player of the Match awards. Filters players with minimum 3 matches, computes composite indices, and serializes clean CSV output.

WHEN Executed when updating the database with new match data or re-engineering feature definitions (`python src/cricket_data_pipeline.py`).

WHY Transforms unstructured granular ball logs into clean, normalized tabular ML vectors.

`data/processed/cricket_players_clean.csv` — Qualified Analytical Dataset

WHAT Structured CSV dataset of 619 qualified professional cricketers across 32 columns (150 KB).

HOW Contains 25 numerical metrics, primary role labels, performance ratings, tier codes, and auction values.

WHEN Loaded by `app.py` during web dashboard runtime and by `train_cricket_models.py` during training.

WHY Lightweight footprint (150 KB) ensures instant git pushes and sub-second container cold-boots on Streamlit Cloud.

`data/raw/matches_2008_2024.csv` — Match Metadata

WHAT Match summaries for all 1,095 IPL matches played from 2008 to 2024 (218 KB).

HOW Contains match dates, venues, participating teams, toss decisions, winners, and Player of the Match awardees.

WHEN Ingested by `cricket_data_pipeline.py` to join match-winning awards.

WHY Player of the Match awards provide empirical evidence of clutch resilience under tournament pressure.

`data/raw/deliveries_2008_2024.csv` — Ball-by-Ball Logs

WHAT Granular ball-by-ball delivery event logs covering 260,920 deliveries across 17 IPL seasons (69 MB).

HOW Contains over-by-over batsman runs, bowler names, dismissals, and extra runs.

WHEN Processed once by the data pipeline script. Ignored by `.gitignore` to keep git repositories fast and lean.

WHY Necessary to compute granular phase-specific metrics (death overs strike rate) not found on standard season scorecards.

2.3 Machine Learning Pipeline & Serialized Models

`src/train_cricket_models.py` — Model Training & Governance Script

WHAT Training script that trains, validates with 5-fold cross-validation, and serializes all 13 ML models.

HOW Executes one-hot encoding, stratified 80/20 train/test split, standard scaling, trains 5 regressors (Linear, Ridge, Poly, RF, MLP), trains 5 classifiers (KNN, RF, Logistic, Decision Tree, MLP), trains K-Means (k=5) and PCA, and dumps all artifacts to `models/` via joblib and JSON.

WHEN Executed during offline model training cycles (`python src/train_cricket_models.py`).

WHY Centralizes model training and governance into a deterministic, reproducible, audited script.

Artifact Path in <code>models/</code>	Format	Serialized Object & Architecture	Role in CricMetrics Pro
<code>models/scaler.joblib</code>	Joblib Binary	StandardScaler(mean_, var_, scale_)	Normalizes 29 raw features to $\mu=0$, $\sigma=1$ during training and inference.
<code>models/random_forest_regressor.joblib</code>	Joblib Binary	RandomForestRegressor(100 trees, max_features='sqrt')	Production Rating Engine: Continuous rating prediction ($R^2 = 0.9789$, RMSE = 1.25).
<code>models/ridge_regression.joblib</code>	Joblib Binary	Ridge(alpha=1.0)	L2 regularized linear regression benchmark for collinear feature stabilization.
<code>models/linear_regression.joblib</code>	Joblib Binary	LinearRegression()	Ordinary Least Squares baseline benchmark model.
<code>models/polynomial_regression.joblib</code>	Joblib Binary	LinearRegression() on PolynomialFeatures(degree=2)	Degree-2 non-linear interaction model capturing skill synergy.
<code>models/mlp_regressor.joblib</code>	Joblib Binary	MLPRegressor(layers=(64, 32), activation='relu')	Deep neural network continuous regression benchmark.
<code>models/knn_classifier.joblib</code>	Joblib Binary	KNeighborsClassifier(k=5, metric='euclidean')	Production Tier Classifier: Categorizes players into talent tiers (95.16% Acc).
<code>models/random_forest_classifier.joblib</code>	Joblib Binary	RandomForestClassifier(100 trees, criterion='gini')	Production ensemble classifier & Gini feature importance engine (94.35% Acc).
<code>models/logistic_regression.joblib</code>	Joblib Binary	LogisticRegression(multinomial, L2)	Multinomial softmax probabilistic classification baseline (87.90% Acc).
<code>models/decision_tree_classifier.joblib</code>	Joblib Binary	DecisionTreeClassifier(max_depth=6, Gini)	White-box rule-based talent classification benchmark (87.90% Acc).
<code>models/mlp_classifier.joblib</code>	Joblib Binary	MLPClassifier(layers=(64, 32), Softmax)	Deep neural network multi-class tier classification benchmark (92.74% Acc).
<code>models/kmeans_model.joblib</code>	Joblib Binary	KMeans(n_clusters=5, init='k-means++')	Unsupervised clustering model discovering 5 tactical playing styles.
<code>models/pca_model.joblib</code>	Joblib Binary	PCA(n_components=2, random_state=42)	Dimensionality reduction engine projecting 29 metrics onto 2D latent map (60.8% var).
<code>models/feature_metadata.json</code>	JSON Text	Feature schemas, column orders, tier mappings	Guarantees exact column ordering and schema consistency between training and app.
<code>models/metrics_summary.json</code>	JSON Text	Benchmark tables, CV scores, confusion matrices	Feeds live leaderboard tables and confusion matrix heatmaps in Tab 6 of app.py.

2.4 Research, Development & Documentation Layer

`notebooks/cricket_player_performance_analysis.ipynb` — Interactive Jupyter Study

WHAT An executed 22-cell data science notebook documenting the complete experimental research study.

HOW Contains interactive cells for data cleaning, correlation heatmaps, model training, 5-fold CV benchmarks, confusion matrices, and PCA plots.

WHEN Executed interactively by data scientists during offline research and peer review.

WHY Provides transparent, step-by-step visual proof of experimental methodologies and results.

`src/create_cricket_notebook.py` — Automated Notebook Generator

WHAT Python utility script programmatically generating `cricket_player_performance_analysis.ipynb` as JSON.

HOW Assembles cells, markdown narratives, and code into standard nbformat v4 structure.

WHEN Executed when research notebook narrative needs automated re-generation.

WHY Guarantees 100% notebook reproducibility without manual editing.

`requirements.txt` — Python Dependency Manifest

WHAT Pinned Python package dependencies manifest (Streamlit, Scikit-Learn, Pandas, Plotly, Joblib).

HOW Read by pip during environment installation and parsed by Streamlit Community Cloud.

WHEN Used during environment setup and container build.

WHY Ensures deterministic, reproducible dependency resolution across workstations and cloud servers.

`.gitignore` — Git Exclusion Rules

WHAT Git exclusion manifest preventing temporary or oversized files from polluting version control.

HOW Ignores `.venv/`, `__pycache__`, and `data/raw/deliveries_*.csv` (69MB).

WHEN Checked by git on every commit and status command.

WHY Prevents hitting GitHub's 100MB file limit while ensuring clean datasets and trained models remain tracked.

`README.md` & `FINAL_REPORT.md` — Executive Documentation

WHAT Public repository landing page and comprehensive formal academic defense dossier.

HOW Contains executive highlights, quickstart commands, formal methodology proofs, and viva voce model answers.

WHEN Read by GitHub visitors, technical auditors, and organizational stakeholders.

WHY Documents the system's compliance with Case Study no. 102 and demonstrates organizational return on investment.

3. End-to-End System Execution Flow Matrix

Component / File	Phase 1: Ingestion & ETL	Phase 2: Training & Serialization	Phase 3: Notebook Research	Phase 4: Cloud Deployment	Phase 5: Executive Handover
data/raw/deliveries_2008_2024.csv	Primary Input	-	-	-	-
data/raw/matches_2008_2024.csv	Primary Input	-	-	-	-
src/cricket_data_pipeline.py	Executes ETL	-	-	-	-
data/processed/cricket_players_clean.csv	Output	Input	Input	Input (Cached)	-
src/train_cricket_models.py	-	Executes Training	-	-	-
models/*.joblib & *.json	-	Outputs	Inputs	Inputs (Cached)	-
notebooks/*.ipynb	-	-	Interactive Execution	-	Research Proof
app.py	-	-	-	Executes App	Live Demo
.streamlit/config.toml	-	-	-	Configures UI	-
requirements.txt	Builds Env	Builds Env	Builds Env	Cloud Build	-
.gitignore	Filters Git	Filters Git	Filters Git	Filters Git	-
README.md	-	-	-	-	Executive Overview
FINAL_REPORT.md	-	-	-	-	Formal Dossier
docs/MACHINE_LEARNING_GUIDE.*	-	-	-	-	ML Encyclopedia
docs/PROJECT_ARCHITECTURE_GUIDE.*	-	-	-	-	System Blueprint